

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2025 P2H Worked Solutions

Jun 2025 | P2H

33 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Statistics Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

- 1** Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 3****TOPIC: Statistics Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate statistic**

The range is 10, and the greatest card is 14:

$$14 - \text{smallest} = 10$$

2. Use Statistics Toolkit tag correct

So the smallest card is 4.

3. Calculate statistic

The median is 7.5. For 6 cards, the median is the mean of the 3rd and 4th cards:

$$\frac{6 + x}{2} = 7.5$$

$$x = 9$$

4. Calculate statistic

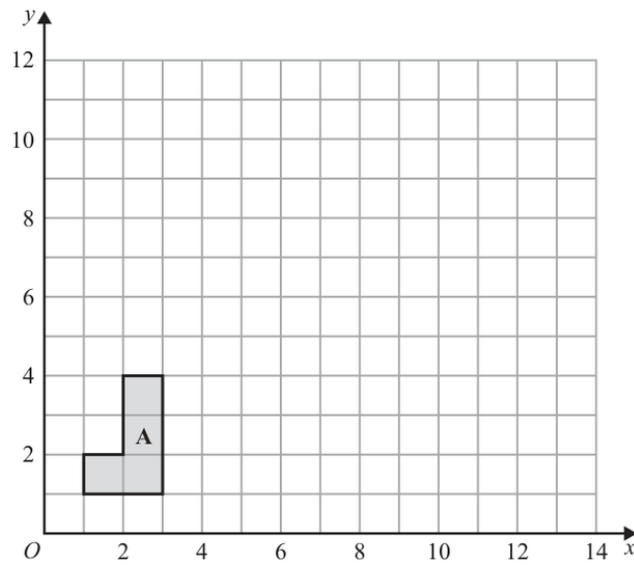
The mode is 6, so the other missing card must be 6.

FINAL ANSWER

4, 6, 9.

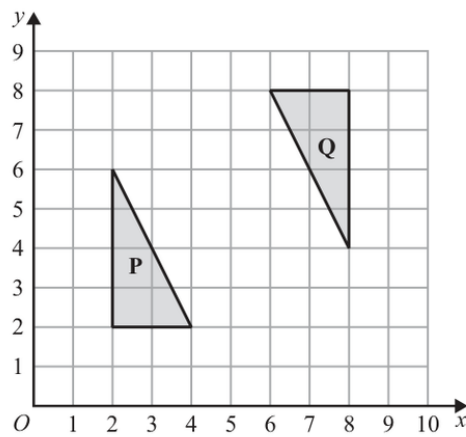
PAST PAPER QUESTION**Q#: 2****Marks: 5****TOPIC: Transformations**

Jun 2025 P2H - Jun 2025 | P2H

2

- (a) On the grid, enlarge shape A with scale factor 3 and centre (0, 1)

(2)



- (b) Describe fully the single transformation that maps triangle P onto triangle Q

(3)

(Total for Question 2 is 5 marks)

PAST PAPER SOLUTION**Q#: 2** **Marks: 5****TOPIC: Transformations**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use Transformations correct**(a) Enlarge from centre $(0, 1)$ with scale factor 3.**2. The vertices become**

The vertices become

 $(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4)$ **3. Use trigonometry**(b) Triangle P is mapped to triangle Q by a 180° rotation about $(5, 5)$.**FINAL ANSWER**(a) vertices $(3, 1), (9, 1), (9, 10), (6, 10), (6, 4), (3, 4)$; (b) rotation 180° about $(5, 5)$.

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

3 Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$

(Total for Question 3 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC: Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$7\frac{1}{3} = \frac{22}{3}, \quad 5\frac{4}{7} = \frac{39}{7}$$

$$7\frac{1}{3} - 5\frac{4}{7} = \frac{22}{3} - \frac{39}{7}$$

$$= \frac{154}{21} - \frac{117}{21} = \frac{37}{21}$$

$$\frac{37}{21} = 1\frac{16}{21}$$

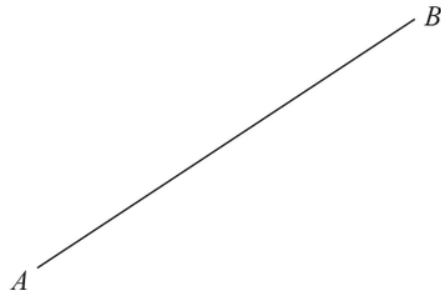
FINAL ANSWER

$$1\frac{16}{21}.$$

PAST PAPER QUESTION**Q#: 4****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

Jun 2025 P2H - Jun 2025 | P2H

- 4 Using ruler and compasses only, construct the perpendicular bisector of the line AB
Show all your construction lines.



(Total for Question 4 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 4 Marks: 2

TOPIC: Bearings, Scale Drawing & Constructions

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Construction answer

To construct the perpendicular bisector of AB :

1. Open the compass to more than half the length of AB .
2. Draw equal arcs from A above and below AB .
3. Without changing the compass width, draw equal arcs from B .
4. Join the two arc intersections with a ruler.

FINAL ANSWER

The joined line is the perpendicular bisector of AB .

EA

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Congruence, Similarity & Geometrical Proof**

Jun 2025 P2H - Jun 2025 | P2H

- 5 The diagram shows two similar quadrilaterals, $ABCD$ and $EFGH$

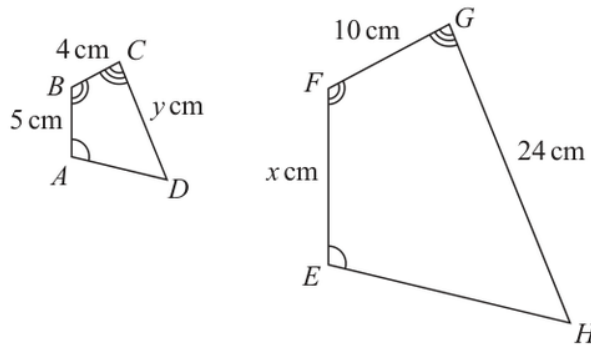


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm} \quad BC = 4 \text{ cm} \quad CD = y \text{ cm}$$

$$EF = x \text{ cm} \quad FG = 10 \text{ cm} \quad GH = 24 \text{ cm}$$

- (a) Work out the value of x

$$x = \dots\dots\dots (2)$$

- (b) Work out the value of y

$$y = \dots\dots\dots (2)$$

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC:** Congruence, Similarity & Geometrical Proof

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**The scale factor from $ABCD$ to $EFGH$ is

$$\frac{FG}{BC} = \frac{10}{4} = 2.5$$

2. Calculate valueSince AB corresponds to EF ,

$$x = EF = 5 \times 2.5 = 12.5$$

3. Calculate valueSince CD corresponds to GH ,

$$2.5y = 24$$

$$y = 9.6$$

FINAL ANSWER

$$x = 12.5, y = 9.6.$$

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Ratio Problem Solving**

Jun 2025 P2H - Jun 2025 | P2H

- 6 Pau, Sam and Tia share £240 in the ratios 3:4:5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.

Give your answer in its simplest form.

..... : :

(Total for Question 6 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 4

TOPIC: **Ratio Problem Solving**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The original ratio is

The original ratio is

$$\text{Pau} : \text{Sam} : \text{Tia} = 3 : 4 : 5$$

2. There are

There are

$$3 + 4 + 5 = 12 \text{ parts}$$

3. Each part is

Each part is

$$240 \div 12 = 20$$

4. Split the ratio

So the original amounts are

$$60, 80, 100$$

5. Sam gives 10 to Pau

Sam gives 10 to Pau and Tia gives 10 to Pau, so the new amounts are

$$80, 70, 90$$

6. Divide by 10

Divide by 10:

$$80 : 70 : 90 = 8 : 7 : 9$$

FINAL ANSWERThe new ratio is $8 : 7 : 9$.

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2025 P2H - Jun 2025 | P2H

- 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.
- Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 7 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 7****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The multiplier for a increase**

The multiplier for a 7% increase is

$$1.07$$

2. Use compound interest

After 3 years,

$$4000(1.07)^3 = 4900.172$$

3. Correct to the nearest Swiss

Correct to the nearest Swiss franc,

$$4900.172 \approx 4900$$

FINAL ANSWER

4900 Swiss francs

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Simultaneous Equations**

Jun 2025 P2H - Jun 2025 | P2H

8 Solve the simultaneous equations

$$\begin{aligned} 3x + 5y &= 8 \\ 4x + y &= -3.5 \end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Simultaneous Equations**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

$$3x + 5y = 8$$

$$4x + y = -3.5$$

2. Solve simultaneous equations

From the second equation,

$$y = -3.5 - 4x$$

3. Substitute into the first equation

Substitute into the first equation:

$$3x + 5(-3.5 - 4x) = 8$$

$$3x - 17.5 - 20x = 8$$

$$-17x = 25.5$$

$$x = -1.5$$

4. Solve simultaneous equations

Then

$$y = -3.5 - 4(-1.5) = 2.5$$

FINAL ANSWER

$$x = -1.5, y = 2.5.$$

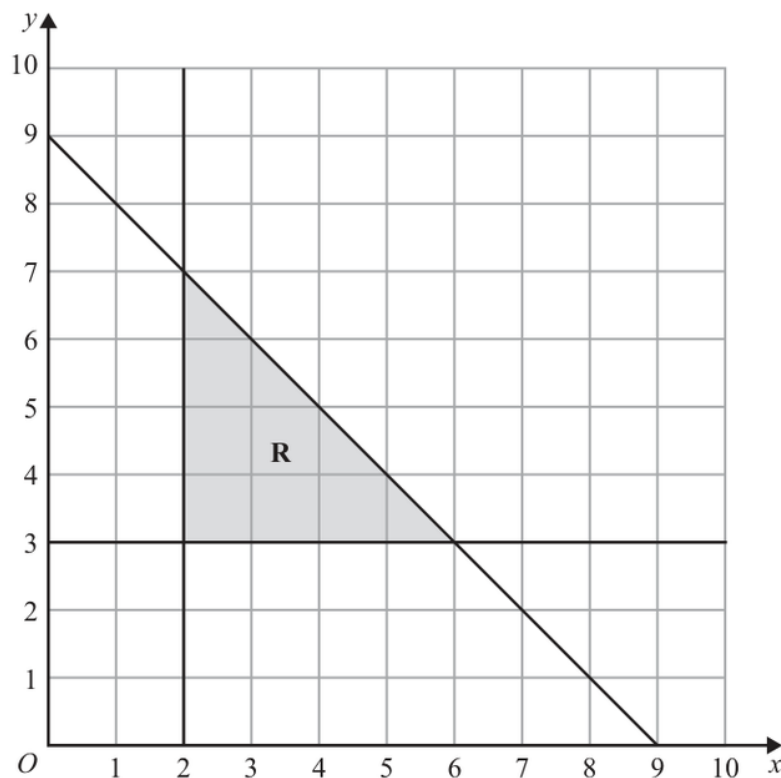
PAST PAPER QUESTION**Q#: 9****Marks: 5****TOPIC: Graphing Inequalities**

Jun 2025 P2H - Jun 2025 | P2H

- 9 (a) Solve the inequality $7 - 3t < 2t + 15$

.....
(2)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



- (b) Write down three inequalities that define the region **R**

.....
(3)

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 5****TOPIC: Graphing Inequalities**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

For part (a),

$$7 - 3x < 2x + 15$$

$$-8 < 5x$$

$$x > -\frac{8}{5}$$

2. Calculate valueFor part (b), the boundaries are $x = 2$, $y = 3$, and $x + y = 9$.**3. The shaded region is**

The shaded region is

$$x \geq 2$$

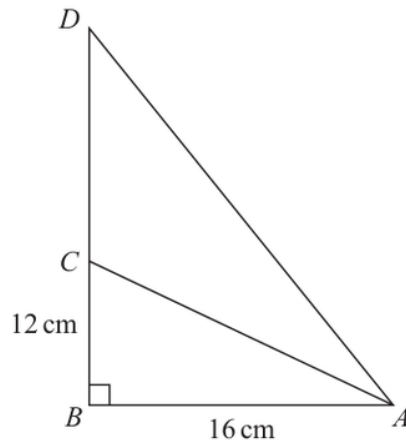
$$y \geq 3$$

$$x + y \leq 9$$

FINAL ANSWER(a) $x > -\frac{8}{5}$. (b) $x \geq 2$, $y \geq 3$, $x + y \leq 9$.

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

10 ABD and ABC are right-angled triangles.Diagram **NOT**
accurately drawn

$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**In triangle ABC ,

$$AC^2 = 16^2 + 12^2 = 400$$

$$AC = 20$$

2. Use trigonometryGiven $AD = 1.5AC$,

$$AD = 30$$

3. Use trigonometryIn triangle ABD ,

$$BD^2 = 30^2 - 16^2 = 644$$

$$BD = 25.377 \dots$$

4. Use trigonometry

So

$$CD = BD - BC = 25.377 \dots - 12 = 13.377 \dots$$

FINAL ANSWER

13.4 cm.

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Probability Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

11 Khalid has a box of counters.

17 of the counters are red

28 of the counters are blue

the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is $\frac{4}{9}$

Work out the number of orange counters that are in the box.

(Total for Question 11 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 11** **Marks: 3****TOPIC: Probability Toolkit**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are counters that are**

There are $17 + 28 = 45$ counters that are not orange.

$$P(\text{orange}) = \frac{4}{9}$$

2. Calculate probability

So

$$P(\text{not orange}) = \frac{5}{9}$$

3. Calculate probability

Let the total number of counters be T .

$$\frac{45}{T} = \frac{5}{9}$$

$$T = 81$$

4. Number of orange counters

Number of orange counters:

$$81 - 45 = 36$$

FINAL ANSWER**36.**

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Expanding Brackets**

Jun 2025 P2H - Jun 2025 | P2H

12 Expand and simplify $3x(2x+5)(7x-4)$

(Total for Question 12 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 3****TOPIC: Expanding Brackets**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Expand brackets**

$$3x(2x + 5)(7x - 4) = 3x(14x^2 + 27x - 20)$$

$$= 42x^3 + 81x^2 - 60x$$

FINAL ANSWER

$$42x^3 + 81x^2 - 60x$$

EA

PAST PAPER QUESTION

Q#: 13 Marks: 6

TOPIC: Cumulative Frequency Diagrams

Jun 2025 P2H - Jun 2025 | P2H

13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

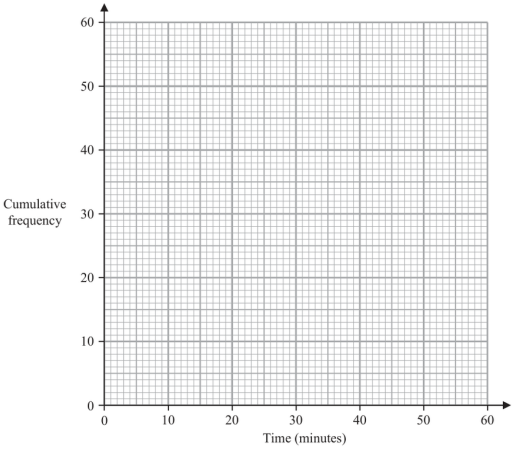
Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	13
$20 < t \leq 30$	12
$30 < t \leq 40$	17
$40 < t \leq 50$	7
$50 < t \leq 60$	3

(a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median time.

_____ minutes
(1)

(d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

(2)

(Total for Question 13 is 6 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Cumulative Frequency Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The cumulative frequencies are**

The cumulative frequencies are

$$8, 21, 33, 50, 57, 60$$

2. Plot

Plot

$$(0, 0), (10, 8), (20, 21), (30, 33), (40, 50), (50, 57), (60, 60)$$

3. The median is the th

The median is the 30th value.

$$20 + \frac{30 - 21}{33 - 21} \times 10 = 27.5$$

4. Use cumulative frequency

At 45 minutes,

$$F(45) \approx 50 + \frac{5}{10}(57 - 50) = 53.5$$

5. Use cumulative frequency

So the number taking more than 45 minutes is

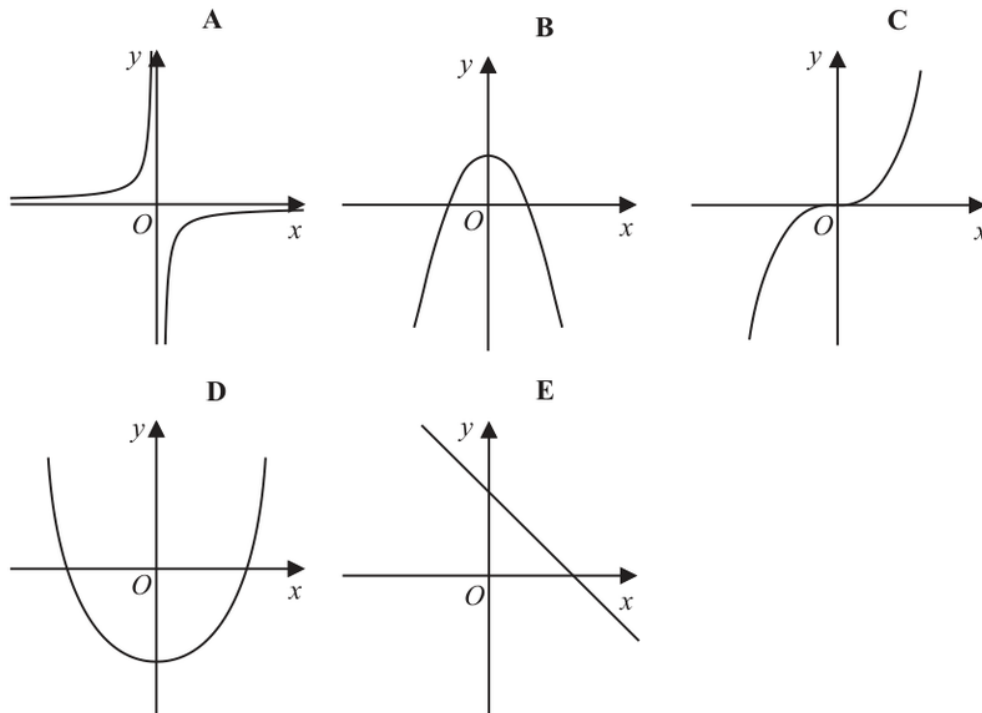
$$60 - 53.5 = 6.5$$

FINAL ANSWER

median about 27.5 minutes, and about 7 people.

PAST PAPER QUESTION**Q#: 14****Marks: 2****TOPIC: Graphs of Functions**

Jun 2025 P2H - Jun 2025 | P2H

14 Here are five graphs.(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$
(1)(b) Write down the letter of the graph that could have the equation $y = 2x^3$
(1)**(Total for Question 14 is 2 marks)**

PAST PAPER SOLUTION**Q#: 14** **Marks: 2****TOPIC: Graphs of Functions**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$y = 5 - x^2$$

2. This is a quadratic graph

This is a quadratic graph with a negative coefficient of x^2 , so it is a downward-opening parabola.

3. That is graph B

That is graph B.

4. (b)

(b)

$$y = 2x^3$$

5. This is an increasing cubic

This is an increasing cubic graph passing through the origin.

6. That is graph C

That is graph C.

FINAL ANSWER

(a) B, (b) C.

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jun 2025 P2H - Jun 2025 | P2H

15 Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

(Total for Question 15 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use index laws**

$$\left(\frac{2a^3}{10a^7c^2} \right)^{-3}$$

2. Use index laws

First simplify inside the bracket:

$$\frac{2a^3}{10a^7c^2} = \frac{1}{5a^4c^2}$$

3. A negative power means take

A negative power means take the reciprocal:

$$\begin{aligned} \left(\frac{1}{5a^4c^2} \right)^{-3} &= (5a^4c^2)^3 \\ &= 125a^{12}c^6 \end{aligned}$$

FINAL ANSWER

$$125a^{12}c^6.$$

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Rearranging Formulas**

Jun 2025 P2H - Jun 2025 | P2H

16 Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$

(Total for Question 16 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Rearranging Formulas**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Rearrange formula**

$$c = \frac{t^2 + 3}{7 - 8t^2}$$

$$c(7 - 8t^2) = t^2 + 3$$

$$7c - 3 = t^2(8c + 1)$$

$$t^2 = \frac{7c - 3}{8c + 1}$$

FINAL ANSWER

$$t = \pm \sqrt{\frac{7c-3}{8c+1}}$$

PAST PAPER QUESTION**Q#: 17****Marks: 6****TOPIC: Differentiation**

Jun 2025 P2H - Jun 2025 | P2H

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$.
Show clear algebraic working.

.....
(4)

.....
(Total for Question 17 is 6 marks)

PAST PAPER SOLUTION**Q#: 17** **Marks: 6****TOPIC: Differentiation**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Differentiate**

$$y = 4x^3 + 5x^2 + 2x$$

$$\frac{dy}{dx} = 12x^2 + 10x + 2$$

2. Differentiate

At a turning point,

$$12x^2 + 10x + 2 = 0$$

$$2(6x^2 + 5x + 1) = 0$$

$$(3x + 1)(2x + 1) = 0$$

$$x = -\frac{1}{3} \quad \text{or} \quad x = -\frac{1}{2}$$

3. DifferentiateWhen $x = -\frac{1}{3}$,

$$y = -\frac{7}{27}$$

4. DifferentiateWhen $x = -\frac{1}{2}$,

$$y = -\frac{1}{4}$$

FINAL ANSWER
 $\left(-\frac{1}{2}, -\frac{1}{4}\right)$ and $\left(-\frac{1}{3}, -\frac{7}{27}\right)$.

PAST PAPER QUESTION**Q#: 18****Marks: 2**TOPIC: **Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 18

Marks: 2

TOPIC: **Algebraic Proof**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let

Let

$$x = 0.\dot{7}\dot{0}\dot{2}$$

2. Calculate value

Then

$$1000x = 702.\dot{7}\dot{0}\dot{2}$$

3. Subtract

Subtract:

$$999x = 702$$

$$x = \frac{702}{999} = \frac{26}{37}$$

FINAL ANSWER

$$0.\dot{7}\dot{0}\dot{2} = \frac{26}{37}$$

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Solving Inequalities**

Jun 2025 P2H - Jun 2025 | P2H

- 19** Solve the inequality $4x^2 + 4x - 15 < 0$
Show clear algebraic working.

(Total for Question 19 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 19** **Marks: 3****TOPIC: Solving Inequalities**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$4x^2 + 4x - 15 < 0$$

2. Factorise

Factorise:

$$(2x + 5)(2x - 3) < 0$$

3. The roots are

The roots are

$$x = -\frac{5}{2}, \quad x = \frac{3}{2}$$

4. The quadratic is negative between

The quadratic is negative between the roots.

FINAL ANSWER

$$-\frac{5}{2} < x < \frac{3}{2}.$$

PAST PAPER QUESTION**Q#: 20****Marks: 6****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

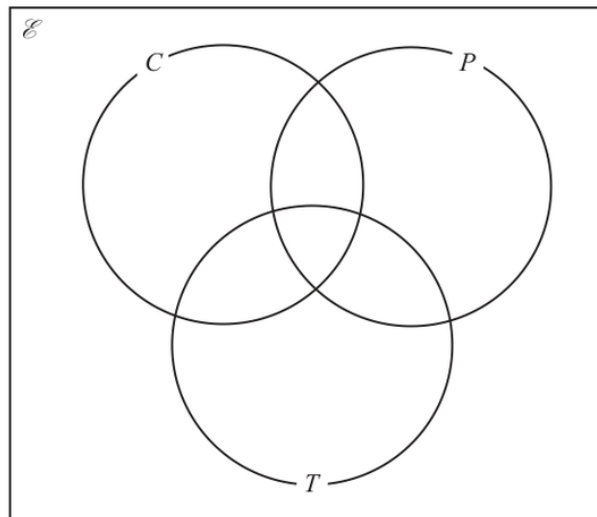
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 6****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

2. The triple intersection is

The triple intersection is

$$n(C \cap P \cap T) = 12$$

3. Calculate value

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

4. Calculate value

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

5. Let the number who grow

Let the number who grow only potatoes be y . The number who grow only tomatoes is also y .

6. Add the regions union

Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

7. Calculate value

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$

8. Calculate value

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

9. Evaluate fraction

For part (c), one gardener who grows carrots is chosen.

$$\begin{aligned} P(\text{also grows potatoes}) &= \frac{n(C \cap P)}{n(C)} \\ &= \frac{18}{43} \end{aligned}$$

FINAL ANSWER

(b) 16, (c) $\frac{18}{43}$

PAST PAPER QUESTION**Q#: 20** **Marks: 6****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

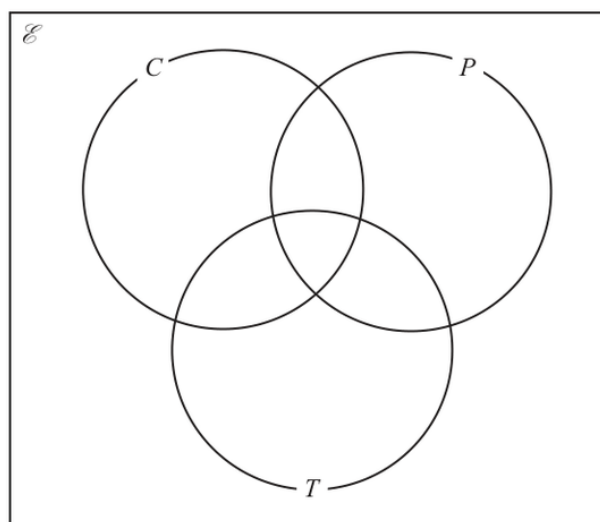
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 6****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

2. The triple intersection is

The triple intersection is

$$n(C \cap P \cap T) = 12$$

3. Calculate value

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

4. Calculate value

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

5. Let the number who grow

Let the number who grow only potatoes be y . The number who grow only tomatoes is also y .

6. Add the regions union

Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

7. Calculate value

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$

8. Calculate value

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

9. Evaluate fraction

For part (c), one gardener who grows carrots is chosen.

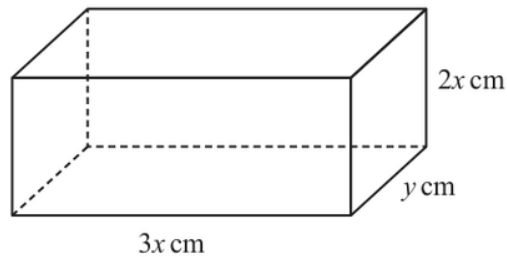
$$\begin{aligned} P(\text{also grows potatoes}) &= \frac{n(C \cap P)}{n(C)} \\ &= \frac{18}{43} \end{aligned}$$

FINAL ANSWER

(b) 16, (c) $\frac{18}{43}$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Volume & Surface Area**

Jun 2025 P2H - Jun 2025 | P2H

21 The diagram shows a cuboid.Diagram **NOT**
accurately drawnThe cuboid measures $3x$ cm by $2x$ cm by y cmThe volume of the cuboid is 1014 cm^3 The total surface area of the cuboid is $A \text{ cm}^2$ Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Volume & Surface Area**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The cuboid has dimensions**

The cuboid has dimensions

$$3x, \quad 2x, \quad y$$

2. Calculate volumeIts volume is 1014 cm^3 , so

$$(3x)(2x)(y) = 1014$$

$$6x^2y = 1014$$

$$y = \frac{1014}{6x^2}$$

$$y = \frac{169}{x^2}$$

3. The total surface area is

The total surface area is

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$A = 2(6x^2 + 5xy)$$

$$A = 12x^2 + 10xy$$

4. SubstituteSubstitute $y = \frac{169}{x^2}$:

$$A = 12x^2 + 10x \left(\frac{169}{x^2} \right)$$

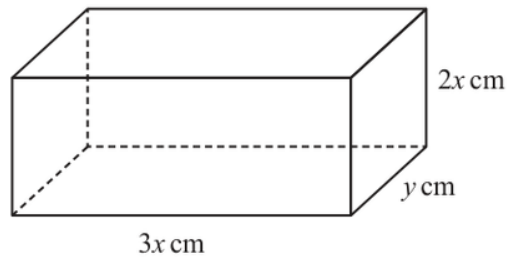
$$A = 12x^2 + \frac{1690}{x}$$

FINAL ANSWER

$$A = 12x^2 + \frac{1690}{x}$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Volume & Surface Area**

Jun 2025 P2H - Jun 2025 | P2H

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(Total for Question 21 is 3 marks)

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Jun 2025 P2H - Jun 2025 | P2H

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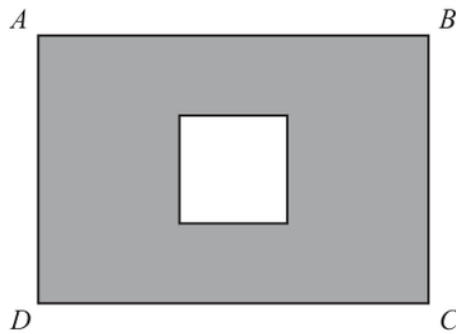
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FINAL ANSWER

$$A = 12x^2 + \frac{1690}{x}$$

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

22 The diagram shows a square inside rectangle $ABCD$ Diagram **NOT**
accurately drawnThe total area of the region shown shaded in the diagram is $X\text{cm}^2$ $AB = 11.5\text{ cm}$ correct to the nearest 0.5 cm $BC = 9.2\text{ cm}$ correct to 2 significant figuresside of square = 4.1 cm correct to 2 significant figuresBy considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly. $X = \dots\dots\dots$ **(Total for Question 22 is 4 marks)**

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The shaded area is

The shaded area is

$$X = \text{area of rectangle} - \text{area of square}$$

2. Bounds

Bounds:

$$11.25 \leq AB < 11.75$$

$$9.15 \leq BC < 9.25$$

$$4.05 \leq \text{square side} < 4.15$$

3. Lower bound

Lower bound:

$$11.25(9.15) - 4.15^2 = 85.715$$

4. Upper bound

Upper bound:

$$11.75(9.25) - 4.05^2 = 92.285$$

5. Estimate the value

So

$$85.715 < X < 92.285$$

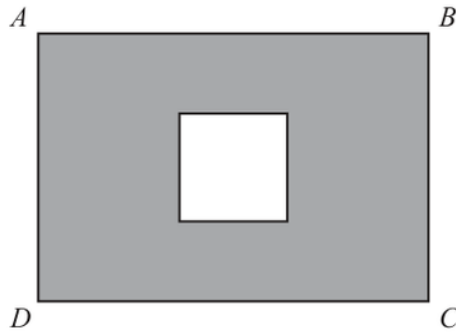
6. Estimate the value

Both bounds round to 90 to the nearest 10.

FINAL ANSWER $X = 90 \text{ cm}^2$ to a suitable degree of accuracy

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

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Show your working clearly. $X = \dots\dots\dots$ **(Total for Question 22 is 4 marks)**

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The shaded area is

The shaded area is

 $X = \text{area of rectangle} - \text{area of square}$ **2. Bounds**

Bounds:

$$11.25 \leq AB < 11.75$$

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$$4.05 \leq \text{side of square} < 4.15$$

3. Find lower boundLower bound for X :

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4. Find upper boundUpper bound for X :

$$11.75(9.25) - 4.05^2 = 92.285$$

5. Estimate the valueSo $85.715 \leq X < 92.285$. Both bounds round to 90 to the nearest 10.**FINAL ANSWER** $X = 90 \text{ cm}^2$ to a suitable degree of accuracy.

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p = \dots\dots\dots$

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**

$$\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x}$$

2. Factorise

Factorise:

$$4x^2 - 4x - 120 = 4(x - 6)(x + 5)$$

$$5x^2 - 180 = 5(x - 6)(x + 6)$$

$$x^2 + 5x = x(x + 5)$$

$$10x^2 + 60x = 10x(x + 6)$$

3. Split the ratio

So

$$\begin{aligned} & \frac{4(x - 6)(x + 5)}{5(x - 6)(x + 6)} \div \frac{x(x + 5)}{10x(x + 6)} \\ &= \frac{4(x + 5)}{5(x + 6)} \times \frac{10(x + 6)}{x + 5} = 8 \end{aligned}$$

FINAL ANSWER

$$p = 8$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

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Find the value of p

Show clear algebraic working.

$p = \dots\dots\dots$

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Algebraic Fractions**

Jun 2025 P2H - Jun 2025 | P2H

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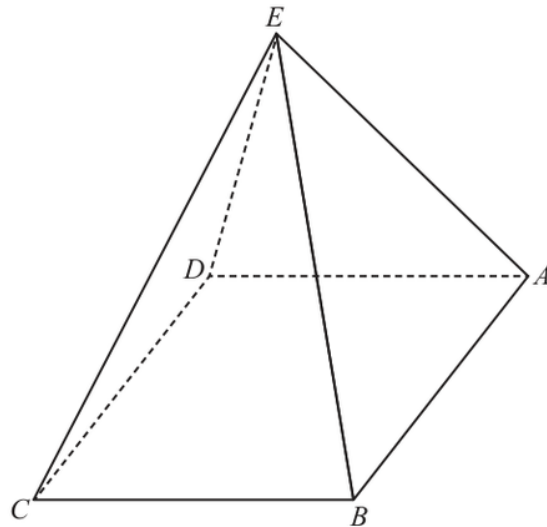
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FINAL ANSWER

$$p = 8$$

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

24 The diagram shows a square-based pyramid $ABCDE$ Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

 M is the centre of the horizontal square base $ABCD$ Q is the midpoint of AB

$$\text{Angle } EQM = 80^\circ$$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let

Let

$$AB = s$$

2. Find the midpointSince M is the centre of the square base and Q is the midpoint of AB ,

$$QM = \frac{s}{2}$$

3. Use trigonometrySince $EA = EB = EC = ED$, the point E is vertically above M .**4. Use trigonometry**In right triangle EQM ,

$$\tan 80^\circ = \frac{EM}{QM}$$

$$EM = \frac{s}{2} \tan 80^\circ$$

5. The distance from the centre

The distance from the centre of a square to a vertex is half the diagonal:

$$AM = \frac{s\sqrt{2}}{2}$$

6. Use trigonometryIn right triangle EMA ,

$$EA^2 = EM^2 + AM^2$$

$$EA^2 = \left(\frac{s}{2} \tan 80^\circ\right)^2 + \left(\frac{s\sqrt{2}}{2}\right)^2$$

$$EA = s\sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

7. Use trigonometry

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

$$n = 2.92$$

FINAL ANSWER

$$n = 2.92$$

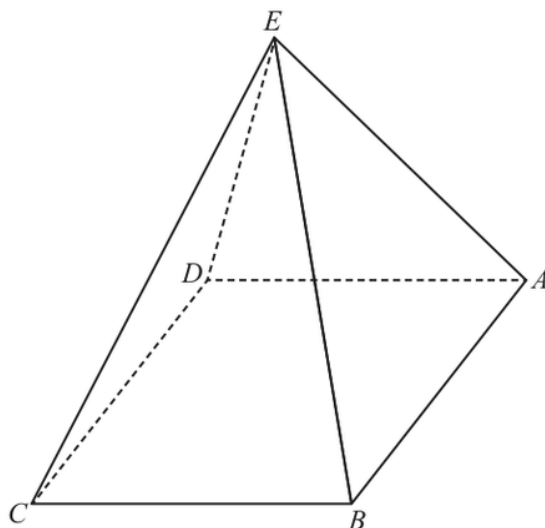
EA

PAST PAPER QUESTION

Q#: 24 Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

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(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jun 2025 P2H - Jun 2025 | P2H

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$$EA = s\sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

7. Use trigonometry

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

$$n = 2.92$$

FINAL ANSWER

$$n = 2.92$$

EA

PAST PAPER QUESTION**Q#: 25****Marks: 6****TOPIC: Completing the Square**

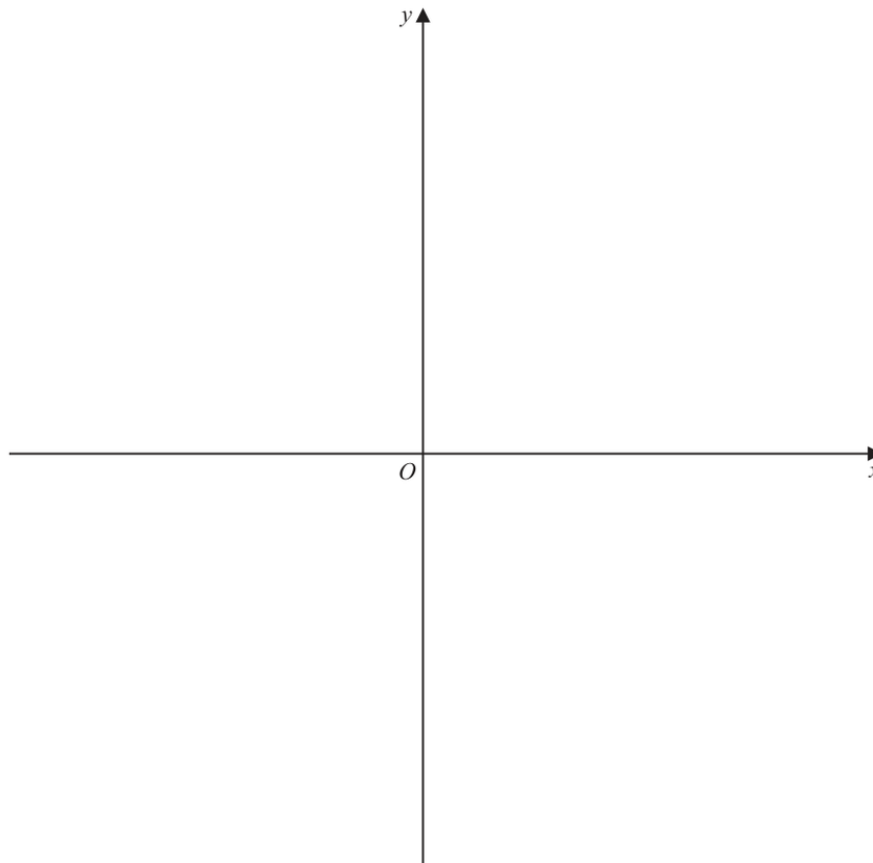
Jun 2025 P2H - Jun 2025 | P2H

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

(3)

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(3)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 6****TOPIC: Completing the Square**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\begin{aligned}
 28 + 24x - 6x^2 &= -6x^2 + 24x + 28 \\
 &= -6(x^2 - 4x) + 28 \\
 &= -6((x - 2)^2 - 4) + 28 \\
 &= 52 - 6(x - 2)^2
 \end{aligned}$$

2. (b)

(b)

$$y = 52 - 6(x - 2)^2$$

3. The graph is a downward

The graph is a downward-opening parabola.

4. The turning point is

The turning point is

$$(2, 52)$$

5. Calculate valueThe point of intersection with the y -axis is found by putting $x = 0$:

$$y = 28$$

6. Find interceptsSo the y -intercept is

$$(0, 28)$$

FINAL ANSWER

(a) $52 - 6(x - 2)^2$. For the sketch, show a downward parabola with turning point $(2, 52)$ and y -intercept $(0, 28)$.

PAST PAPER QUESTION**Q#: 25****Marks: 6****TOPIC: Completing the Square**

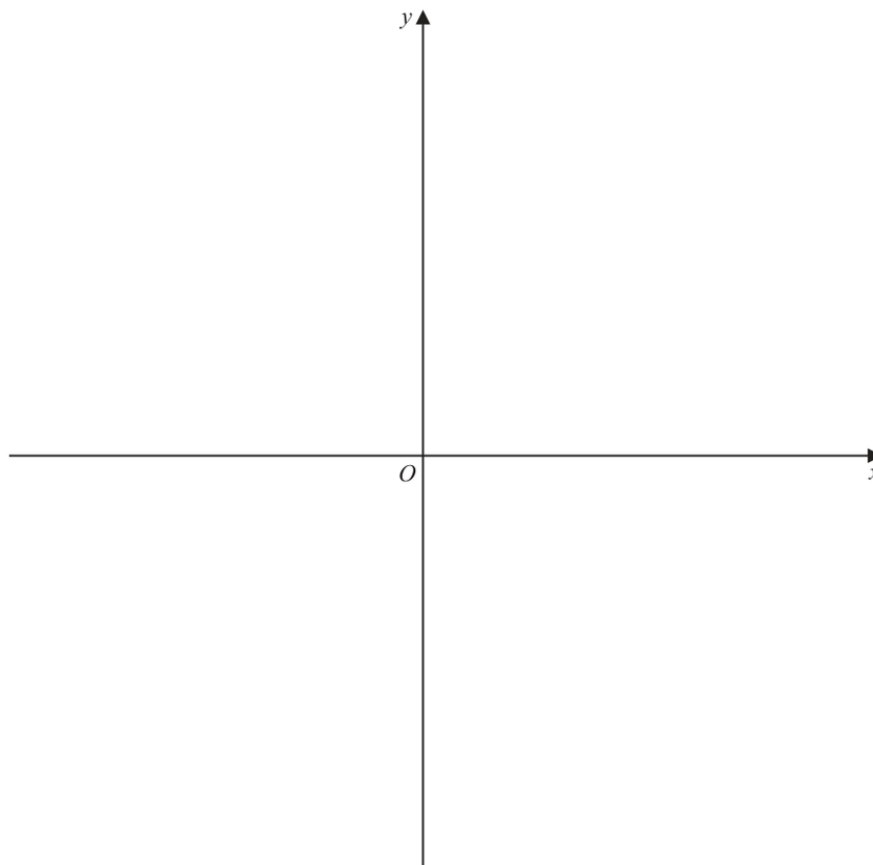
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(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 6****TOPIC: Completing the Square**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

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FINAL ANSWER

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PAST PAPER QUESTION**Q#: 26****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

26 **A** and **B** are two similar solids.The height of solid **A** is 31 cmThe height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**..... cm³**(Total for Question 26 is 4 marks)**

EA

PAST PAPER SOLUTION**Q#: 26****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The heights of solids A**

The heights of solids A and B are 31 cm and 18.6 cm.

$$31 : 18.6 = 5 : 3$$

2. Split the ratio

So the volumes are in the ratio

$$5^3 : 3^3 = 125 : 27$$

3. The difference in volume is

The difference in volume is

$$125 - 27 = 98$$

4. parts

parts.

5. Evaluate fractionThis difference is 735 cm^3 , so one part is

$$\frac{735}{98} = 7.5$$

6. Volume of solid A

Volume of solid A:

$$125 \times 7.5 = 937.5$$

FINAL ANSWER

$$937.5 \text{ cm}^3$$

PAST PAPER QUESTION**Q#: 26****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

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The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 26

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P2H - Jun 2025 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

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